

Flow Past a Cylinder

Introduction

The flow of fluid behind a blunt body such as an automobile is difficult to compute due to the unsteady flows. The wake behind such a body consists of unordered eddies of all sizes that create large drag on the body. In contrast, the turbulence in the thin boundary layers next to the streamlined bodies of aircraft and fish create only weak disturbances of flow.

An exception to this occurs when you place a slender body at right angles to a slow flow because the eddies organize. A von Kármán vortex street appears with a predictable frequency and involves the shedding of eddies from alternating sides. Everyday examples of this phenomenon include singing telephone wires and an automobile radio antenna vibrating in an air stream.

From an engineering standpoint, it is important to predict the frequency of vibrations at various fluid speeds and thereby avoid undesirable resonances between the vibrations of the solid structures and the vortex shedding. To help reduce such effects, plant engineers put a spiral on the upper part of high smokestacks; the resulting variation in shape prohibits the constructive interference of the vortex elements that the structure sheds from different positions.

Model Definition

To illustrate how you can study such effects, the following model examines unsteady, incompressible flow past a long cylinder placed in a channel at right angle to the oncoming fluid. With a symmetric inlet velocity profile, the flow needs some kind of asymmetry to trigger the vortex production. This can be achieved by placing the cylinder with a small offset from the center of the flow. In this case, an unstructured mesh is used, and the small asymmetry in the mesh proves to be enough to trigger the vortex production.

The simulation time necessary for a periodic flow pattern to appear is difficult to predict. A key predictor is the Reynolds number, which is based on cylinder diameter. For low values (below 100) the flow is steady. In this simulation, the Reynolds number equals 100, which gives a developed von Kármán vortex street, but the flow still is not fully turbulent.

The frequency and amplitude of oscillations are stable features, but flow details are extremely sensitive to perturbations. To gain an appreciation for this sensitivity, you can compare flow images taken at the same time but with such minor differences as are created by different tolerances for the time stepping. It is important to note that this sensitivity is a physical reality and not simply a numerical artifact.

Before calculating the time-varying forces on the cylinder, you can validate the method of computation at a lower Reynolds number using the direct nonlinear solver. This saves time because you can find and correct simple errors and mistakes before the final time-dependent simulation, which requires considerable time.

The viscous forces on the cylinder are proportional to the gradient of the velocity field at the cylinder surface. Evaluating the velocity gradient on the boundary by directly differentiating the FEM solution is possible but not very accurate. The differentiation produces 1st-order polynomials when second-order elements are used for the velocity field. A far better approach is to use a pair of reaction force operators to compute the integrals of the viscous forces, comparable to second-order accurate integrals of the viscous forces. An alternative approach would be to use a pair of weak-constraint variables to enforce the no slip condition. Preferably use the reaction force operator instead of weak constraints when computing integrals of reaction forces or fluxes in postprocessing.

The drag and lift forces themselves are not as interesting as the dimensionless drag and lift coefficients. These depend only on the Reynolds number and an object's shape, not its size. The coefficients are defined as

$$C_D = \frac{2F_D}{\rho U_{\text{mean}}^2 L}$$
$$C_L = \frac{2F_L}{\rho U_{\text{mean}}^2 L}$$

using the following parameters:

- F_D and F_L are the drag and lift forces
- ρ is the fluid's density
- U_{mean} is the mean velocity
- L is the characteristic length, in this case the cylinder's radius

Results and Discussion

Figure 1 shows the flow pattern resulting from the geometry.

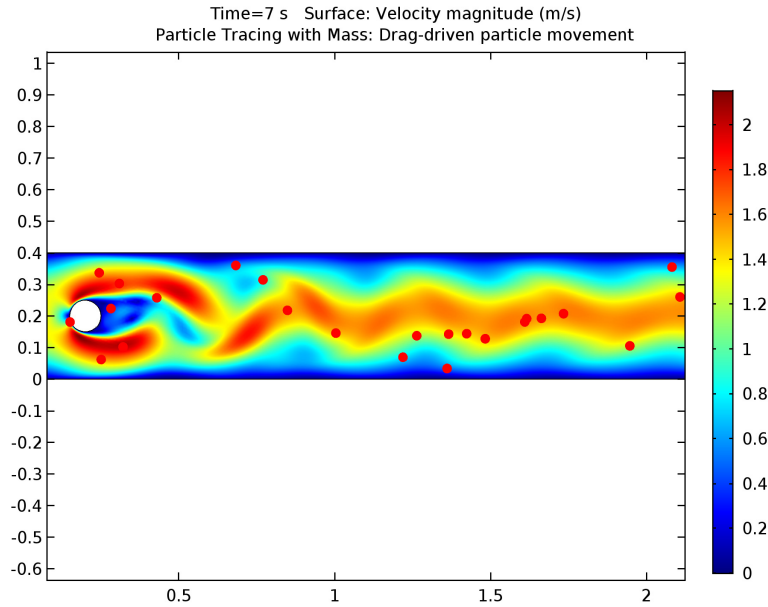


Figure 1: A plot of the last time step clearly shows the von Kármán path.

The flow around a cylinder is a common benchmark test for CFD algorithms. Various research teams have tried their strengths on this problem using different techniques. Results from some of these experiments have been collected by Schäfer and Turek (Ref. 1), who also used them to compute a probable value for the “real” answer.

Figure 2 shows how the lift coefficient develops a periodic variation as the von Kármán vortex structure is formed.

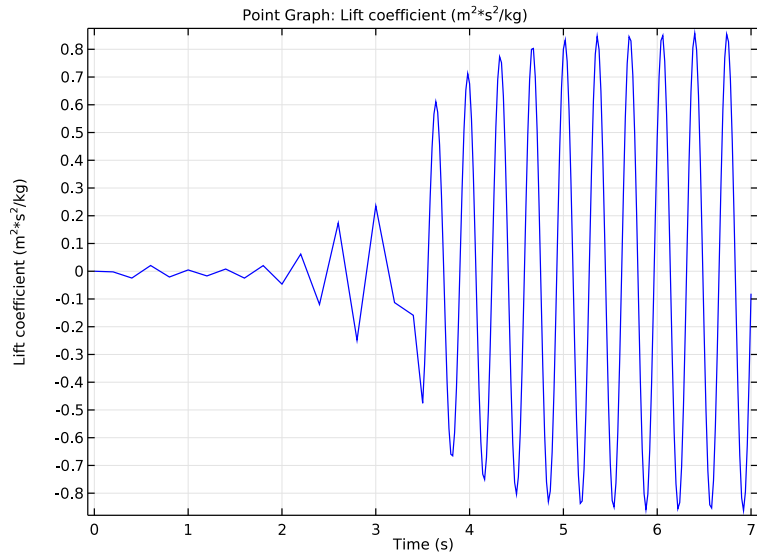


Figure 2: Lift coefficient, C_L , as a function of time.

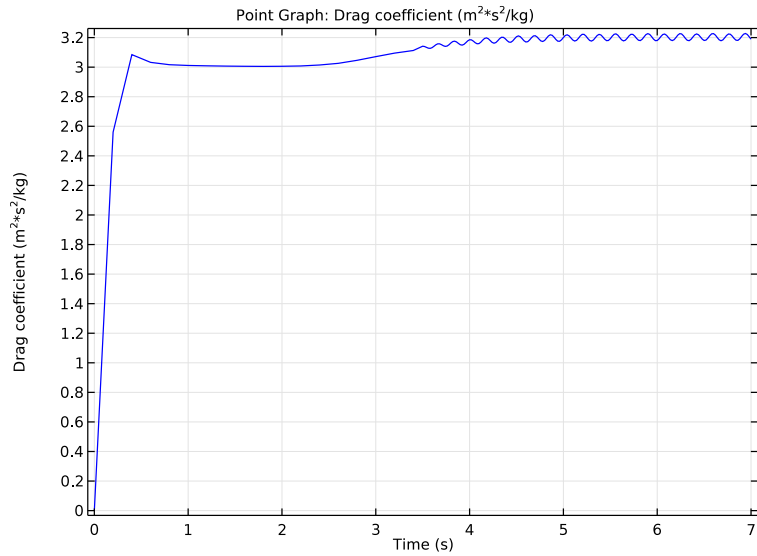


Figure 3: Drag coefficient, C_D , as a function of time.

Reference

1. M. Schäfer and S. Turek, “Benchmark Computations of Laminar Flow Around Cylinder,” E.H. Hirschel ed., *Flow Simulation with High-Performance Computers II, Volume 52 of Notes on Numerical Fluid Mechanics*, Vieweg, pp. 547–566, 1996.

Application Library path: COMSOL_Multiphysics/Fluid_Dynamics/cylinder_flow

Modeling Instructions

From the **File** menu, choose **New**.

NEW

In the **New** window, click **Model Wizard**.

MODEL WIZARD

- 1 In the **Model Wizard** window, click **2D**.
- 2 In the **Select Physics** tree, select **Fluid Flow>Single-Phase Flow>Laminar Flow (spf)**.
- 3 Click **Add**.
- 4 Click **Study**.
- 5 In the **Select Study** tree, select **Preset Studies>Time Dependent**.
- 6 Click **Done**.

GLOBAL DEFINITIONS

Parameters

- 1 On the **Home** toolbar, click **Parameters**.
- 2 In the **Settings** window for Parameters, locate the **Parameters** section.
- 3 In the table, enter the following settings:

Name	Expression	Value	Description
U_mean	1[m/s]	1 m/s	Mean inflow velocity

Next, create a smoothed step function feature that you will use for ramping up the inflow velocity.

Step 1 (step1)

- 1 On the **Home** toolbar, click **Functions** and choose **Global>Step**.
- 2 In the **Settings** window for Step, locate the **Parameters** section.
- 3 In the **Location** text field, type 0.1.

GEOMETRY 1

Rectangle 1 (r1)

- 1 On the **Geometry** toolbar, click **Primitives** and choose **Rectangle**.
- 2 In the **Settings** window for Rectangle, locate the **Size and Shape** section.
- 3 In the **Width** text field, type 2.2.
- 4 In the **Height** text field, type 0.4.

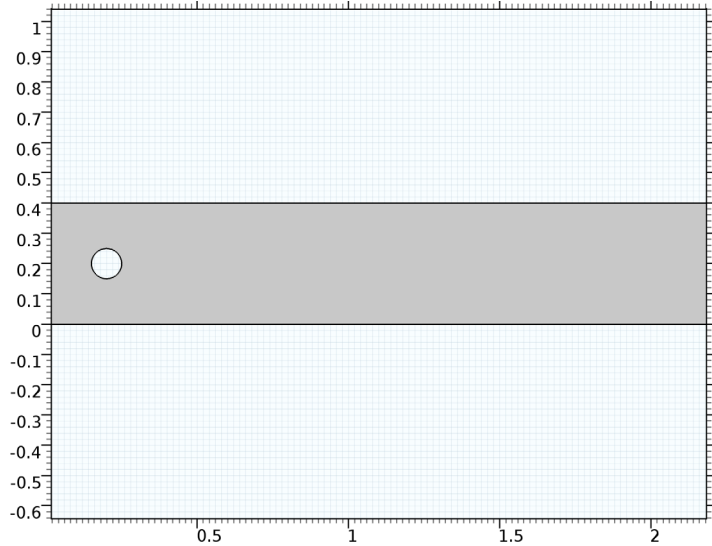
Circle 1 (c1)

- 1 Right-click **Rectangle 1 (r1)** and choose **Build Selected**.
- 2 On the **Geometry** toolbar, click **Primitives** and choose **Circle**.
- 3 In the **Settings** window for Circle, locate the **Position** section.
- 4 In the **x** text field, type 0.2.
- 5 In the **y** text field, type 0.2.
- 6 Locate the **Size and Shape** section. In the **Radius** text field, type 0.05.

Difference 1 (dif1)

- 1 Right-click **Circle 1 (c1)** and choose **Build Selected**.
- 2 On the **Geometry** toolbar, click **Booleans and Partitions** and choose **Difference**.
- 3 Select the object **r1** only.
- 4 In the **Settings** window for Difference, locate the **Difference** section.
- 5 Find the **Objects to subtract** subsection. Select the **Active** toggle button.
- 6 Select the object **c1** only.
- 7 On the **Geometry** toolbar, click **Build All**.

8 Click the **Zoom Extents** button on the **Graphics** toolbar.



MATERIALS

In the **Model Builder** window, under **Component 1 (comp1)** right-click **Materials** and choose **Blank Material**.

Material 1 (mat1)

- 1 In the **Settings** window for Material, locate the **Material Contents** section.
- 2 In the table, enter the following settings:

Property	Name	Value	Unit	Property group
Density	rho	1	kg/m ³	Basic
Dynamic viscosity	mu	1 e -3	Pa·s	Basic

LAMINAR FLOW (SPF)

Inlet 1

- 1 On the **Physics** toolbar, click **Boundaries** and choose **Inlet**.
- 2 Select Boundary 1 only.

To define a parabolic velocity profile use the predefined local curve parameter *s*. Ramp up the velocity using the previously defined step function. Append the inverse unit bracket [1/s] to the time variable *t* because the step function expects a dimensionless argument.

- 3 In the **Settings** window for Inlet, locate the **Velocity** section.
- 4 In the U_0 text field, type $U_{\text{mean}}*6*s*(1-s)*\text{step1}(t[1/s])$.

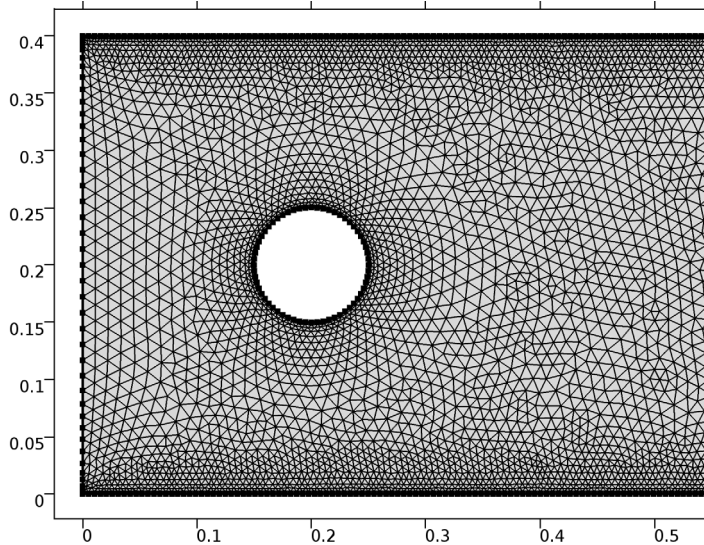
Outlet 1

- 1 On the **Physics** toolbar, click **Boundaries** and choose **Outlet**.
- 2 Select Boundary 4 only.

MESH 1

- 1 In the **Model Builder** window, under **Component 1 (comp1)** click **Mesh 1**.
- 2 In the **Settings** window for Mesh, locate the **Mesh Settings** section.
- 3 From the **Element size** list, choose **Finer**.
- 4 Click **Build All**.

If you zoom in on the inlet and the cylinder you can see the boundary layers that the physics-controlled mesh gives.



STUDY 1

Step 1: Time Dependent

- 1 In the **Model Builder** window, expand the **Study 1** node, then click **Step 1: Time Dependent**.
- 2 In the **Settings** window for Time Dependent, locate the **Study Settings** section.
- 3 In the **Times** text field, type $\text{range}(0,0.2,3.4) \text{ range}(3.5,0.02,7)$.

Solution 1 (sol1)

- 1 On the **Study** toolbar, click **Show Default Solver**.
- 2 In the **Model Builder** window, expand the **Study 1>Solver Configurations** node.
- 3 In the **Model Builder** window, expand the **Solution 1 (sol1)** node, then click **Time-Dependent Solver 1**.
- 4 In the **Settings** window for Time-Dependent Solver, click to expand the **Time stepping** section.
- 5 Locate the **Time Stepping** section. From the **Steps taken by solver** list, choose **Intermediate**.
- 6 On the **Study** toolbar, click **Compute**.

RESULTS

Velocity (spf)

Add a **Particle Tracing with Mass** node to the first default plot group to reproduce the plot in [Figure 1](#).

- 1 In the **Model Builder** window, click **Velocity (spf)**.

Particle Tracing with Mass 1

On the **Velocity (spf)** toolbar, click **More Plots** and choose **Particle Tracing with Mass**.

Velocity (spf)

- 1 In the **Settings** window for Particle Tracing with Mass, click to expand the **Mass and velocity** section.
- 2 Locate the **Mass and Velocity** section. In the **Mass** text field, type $4\pi/3 \cdot 1e-9$.
- 3 Find the **Initial velocity** subsection. In the **x component** text field, type u .
- 4 In the **y component** text field, type v .
- 5 Locate the **Particle Positioning** section. In the **y** text field, type $\text{range}(0.1, 0.05, 0.3)$.
- 6 In the **x** text field, type 0 .
- 7 Click to expand the **Release** section. From the **Release particles** list, choose **At intervals**.
- 8 Select the **Start time** check box.
- 9 In the associated text field, type 3.6 .
- 10 In the **Time between releases** text field, type 0.4 .
- 11 Click to expand the **Coloring and style** section. Locate the **Coloring and Style** section. Find the **Line style** subsection. From the **Type** list, choose **None**.
- 12 Find the **Point style** subsection. From the **Type** list, choose **Point**.

13 Find the **Point motion** subsection. From the **When particle leaves domain** list, choose **Disappear**.

14 On the **Velocity (spf)** toolbar, click **Plot**.

To reproduce [Figure 2](#) and [Figure 3](#) of the lift and drag coefficients, first add an Integral data set for computing the total reaction force on the cylinder.

Integral 1

On the **Results** toolbar, click **More Data Sets** and choose **Evaluation>Integral**.

Selection

1 On the **Results** toolbar, click **Selection**.

2 In the **Settings** window for Selection, locate the **Geometric Entity Selection** section.

3 From the **Geometric entity level** list, choose **Boundary**.

4 Select Boundaries 5–8 only.

1D Plot Group 3

1 On the **Results** toolbar, click **1D Plot Group**.

2 In the **Settings** window for 1D Plot Group, locate the **Data** section.

3 From the **Data set** list, choose **Integral 1**.

Point Graph 1

On the **1D Plot Group 3** toolbar, click **Point Graph**.

1D Plot Group 3

1 In the **Settings** window for Point Graph, locate the **y-Axis Data** section.

2 In the **Expression** text field, type $-reacf(v)^2/(spf.rho*U_mean^2*0.1)$.

3 Select the **Description** check box.

4 In the associated text field, type **Lift coefficient**.

5 On the **1D Plot Group 3** toolbar, click **Plot**.

Compare the graph with that shown in [Figure 2](#).

Finally, visualize the drag coefficient using the following steps:

1D Plot Group 4

1 On the **Home** toolbar, click **Add Plot Group** and choose **1D Plot Group**.

2 In the **Settings** window for 1D Plot Group, locate the **Data** section.

3 From the **Data set** list, choose **Integral 1**.

Point Graph 1

On the **ID Plot Group 4** toolbar, click **Point Graph**.

ID Plot Group 4

- 1** In the **Settings** window for Point Graph, locate the **y-Axis Data** section.
- 2** In the **Expression** text field, type $-reacf(u)^2/(spf.rho*U_mean^2*0.1)$.
- 3** Select the **Description** check box.
- 4** In the associated text field, type Drag coefficient.
- 5** On the **ID Plot Group 4** toolbar, click **Plot**.

Compare with [Figure 3](#).